Shuvomoy Das Gupta

CONTACT

6100 Main St MS-134, Houston, TX 77005, USA
https://shuvomoy.github.io/
sd158@rice.edu

CITIZENSHIP Canada

RESEARCH INTERESTS Optimization, Game Theory, Computation, Transportation

CURRENT POSITION

Rice University, Houston, TX, USA 2025–Present
Assistant Professor, Computational Applied Mathematics & Operations Research

ACADEMIC EXPERIENCE

Columbia University, New York, NY, USA 2024–2025
Postdoctoral Research Scientist, Industrial Engineering & Operations Research
HOSTS: Garud Iyengar & Christian Kroer
Worked on designing optimal algorithms for large-scale game solving.

INDUSTRY EXPERIENCE

Thales Canada Inc., Toronto, Canada 2016–2018
Researcher, Research & Technology Department
Worked on real-time embedded optimization and sensor fusion algorithms in autonomous transportation systems.

EDUCATION

Massachusetts Institute of Technology, Cambridge, USA 2019 – 2024
Ph.D. in Operations Research
THESIS: Advances in Computer-Assisted Design and Analysis of First-Order Optimization Methods and Related Problems
PDF: https://dspace.mit.edu/handle/1721.1/155495
ADVISORS: Robert M. Freund & Bart P.G. Van Parys

University of Toronto, Toronto, Canada 2016
Master of Applied Science in Electrical and Computer Engineering
THESIS: Optimization Models for Energy-Efficient Railway Timetables
PDF: https://utoronto.scholaris.ca/items/b89b9f6b-f9db-4757-8971-6d1563f779fd
ADVISOR: Lacra Pavel

GRANTS, AWARDS AND HONORS

AFOSR Grant: "Computer-Assisted Design of Provably Fastest Algorithms". Co-PI. $600,000. (my share: $300,000) 2025–2028

Winner, INFORMS Computing Society Student Paper Award 2024

Honorable Mention & Finalist, INFORMS Nicholson Student Paper Competition 2024

Honorable Mention, MIT Operations Research Center Best Student Paper Award 2024

PUBLISHED JOURNAL PAPERS

Uijeong Jang, Shuvomoy Das Gupta, and Ernest K. Ryu, "**Computer-Assisted Design of Accelerated Composite Optimization Methods: OptISTA**," published in *Mathematical Programming Series A*, 2025.
PDF: `https://arxiv.org/pdf/2305.15704.pdf`

Shuvomoy Das Gupta, Bart P.G. Van Parys, and Ernest K. Ryu, "**Branch-and-Bound Performance Estimation Programming: A Unified Methodology for Constructing Optimal Optimization Methods**," published in *Mathematical Programming Series A*, 2024.
PDF: `https://arxiv.org/pdf/2203.07305.pdf`

Shuvomoy Das Gupta, Robert M. Freund, Xu Andy Sun, Adrien B. Taylor, "**Nonlinear Conjugate Gradient Methods: Worst-case Convergence Rates via Computer-assisted Analyses**," published in *Mathematical Programming Series A*, 2024.
PDF: `https://arxiv.org/pdf/2301.01530.pdf`

Shuvomoy Das Gupta, Bartolomeo Stellato, and Bart P.G. Van Parys, "**Exterior-point Optimization for Sparse and Low-rank Optimization**," published in the *Journal of Optimization Theory and Applications*, 2024.
PDF: `https://arxiv.org/pdf/2011.04552.pdf`

Shuvomoy Das Gupta and Lacra Pavel, "**On Seeking Efficient Pareto Optimal Points in Multi-Player Minimum Cost Flow Problems with Application to Transportation Systems**," published in the *Journal of Global Optimization* 74 (2019): 523–548.
PDF: `https://arxiv.org/pdf/1805.11750.pdf`

Shuvomoy Das Gupta, J. Kevin Tobin, and Lacra Pavel, "**A Two-Step Linear Programming Model for Energy-Efficient Timetables in Metro Railway Networks**," published in *Transportation Research Part B: Methodological* 93 (2016): 57–74.
PDF: `https://arxiv.org/pdf/1506.08243.pdf`

PUBLISHED CONFERENCE PAPERS

Tianlong Nan, Shuvomoy Das Gupta, Garud Iyengar, Christian Kroer, "**On the $O(1/T)$ Convergence of Alternating Gradient Descent–Ascent in Bilinear Games**," published in *ICLR*, 2026.
PDF: `https://arxiv.org/pdf/2510.03855`

Jakub Cerny, Shuvomoy Das Gupta, and Christian Kroer, "**Spatial Branch-and-Bound for Computing Multiplayer Nash Equilibrium**," published in *AAAI*, 2026.
PDF: `https://arxiv.org/pdf/2508.10204`

Shuvomoy Das Gupta, "**On Convergence of Heuristics Based on Douglas-Rachford Splitting and ADMM to Minimize Convex Functions over Nonconvex Sets**," published in the *proceedings of the 56th Allerton Conference on Communication, Control, and Computing*, University of Illinois at Urbana-Champaign, IL, USA, October 2018.
PDF: `https://shuvomoy.github.io/Papers/Allerton_2018.pdf`

Shuvomoy Das Gupta and Lacra Pavel, "**Multi-player Minimum Cost Flow Problems with Nonconvex Costs and Integer Flows**," published in the *proceedings of the 55th IEEE Conference on Decision and Control*, Las Vegas, NV, USA, December 12–14, 2016.
PDF: `https://shuvomoy.github.io/Papers/Multi-player_minimum_cost_flow_problems_with_nonconvex_costs_and_integer_flows.pdf`

Shuvomoy Das Gupta, Lacra Pavel, and J. Kevin Tobin, "**An Optimization Model to Utilize Regenerative Braking Energy in a Railway Network**," published in the *proceedings of 2015 American Control Conference (ACC)*, Chicago, IL, USA, July 2015.
PDF: `https://shuvomoy.github.io/Papers/An_Optimization_Model_to_Utilize_Regenerative_Braking_Energy_in_a_Railway_Network.pdf`

Preprints

Manu Upadhyaya, Shuvomoy Das Gupta, Adrien B. Taylor, Sebastian Banert, and Pontus Giselsson, "**The AutoLyap Software Suite for Computer-Assisted Lyapunov Analyses of First-Order Methods**," arXiv preprint, 2026.
PDF: `https://arxiv.org/pdf/2506.24076`

Teaching

CMOR 537: Computer-assisted Algorithm Design for Optimization and Machine Learning, Rice — Spring 2026
Instructor. New graduate course for its inaugural offering.

CMOR 467/567: Optimization for Energy Systems, Rice — Fall 2025
Instructor. I have designed and launched this new course for advanced undergraduate and graduate students.

6.7220: Nonlinear Optimization, MIT — Spring 2023
Teaching Assistant. This is MIT's main doctoral course in optimization.
RATING: 6.9/7.0

15.S60: Computing in Optimization and Statistics, MIT — Winter 2023
Instructor. I taught the ORC's required three-hour module on advanced methods in computational optimization.
RATING: 7/7

15.S60: Computing in Optimization and Statistics, MIT — Winter 2022
Instructor. I taught the ORC's required three-hour module on advanced methods in computational optimization.
RATING: 6.8/7

15.S08: Optimization of Energy Systems, MIT — Spring 2022
Teaching Assistant. This is a graduate course in power systems modeling and optimization.
RATING: 6.0/7.0

Talks

On the $O(1/T)$ Convergence of Alternating Gradient Descent-Ascent in Bilinear Games
INFORMS Annual Meeting, Atlanta, GA — 2025

Computer-Assisted Design of Provably Fastest Algorithms
Invited talk, New Jersey Institute of Technology, New York, NY — 2025

Nonlinear Conjugate Gradient Methods: Worst-case Convergence Rates via Computer-assisted Analyses
ICCOPT, Los Angeles, CA 2025
INFORMS Annual Meeting, Seattle, WA 2024

BnB-PEP: A Unified Methodology for Constructing Optimal Optimization Methods
INFORMS Annual Meeting, Phoenix, AZ 2023
SIAM Conference on Optimization (OP23), Seattle, Washington 2023
UTORG Seminar, University of Toronto, Toronto, Canada 2023
International Conference on Continuous Optimization, Bethlehem, PA 2022
MIT Data Science Lab Seminar 2022

Design and Analysis of First-Order Methods via Nonconvex QCQP Frameworks
One of just four invited "long talks" at the 1st Workshop on Performance Estimation, UCLouvain, Belgium 2023

Energy-Optimal Timetable Design for Sustainable Metro Railway Networks
INFORMS Annual Meeting, Phoenix, AZ 2023
33rd Annual POMS Conference, Orlando, FL 2023
2023 MIT Energy Initiative Annual Research Conference 2023

NExOS.jl for Nonconvex Exterior-point Optimization
JuliaCon 2021

Exterior-point Optimization for Sparse and Low-rank Optimization
Seoul National University 2020
INFORMS Annual Meeting (virtual) 2020
MIT LIDS student conference 2021

On Convergence of Heuristics Based on Douglas-Rachford Splitting and ADMM to Minimize Convex Functions over Nonconvex Sets
56th Allerton Conference on Communication, Control, and Computing, Monticello, IL 2018

Multi-Player Minimum Cost Flow Problems with Nonconvex Costs and Integer Flows
55th IEEE Conference on Decision and Control, Las Vegas, NV 2016

On seeking efficient Pareto optimal points in multi-player minimum cost flow problems
MIT LIDS student conference 2020

Service

Organizer, Friends of Optimization Seminar Series 2024–
LINK: `https://sites.google.com/view/friends-of-optimization`

Reviewer for *Mathematical Programming, Management Science, Transportation Research Part B: Methodological, IEEE Transactions on Control of Network Systems, American Control Conference, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Automatic Control*

Session Chair, INFORMS Annual Meeting 2022–2025

Software

[1] PEPit.jl
Native Julia implementation of the Performance Estimation Programming (PEP) methodology.
LINK: `https://github.com/PerformanceEstimation/PEPit.jl`

[2] AutoLyap.jl
Native Julia implementation of the AutoLyap methodology.
LINK: `https://github.com/AutoLyap/AutoLyap.jl`

[3] BnB-PEP
Computes optimal first-order algorithms for different convex and nonconvex setups
LINK: `https://github.com/Shuvomoy/BnB-PEP-code`

[4] NCG-PEP
Computes worst-case convergence rates of nonlinear conjugate gradient methods
LINK: `https://github.com/Shuvomoy/NCG-PEP-code`

[5] NExOS
Implements the Nonconvex Exterior-point Optimization Solver (NExOS) algorithm for solving low-rank and sparse optimization problems
LINK: `https://github.com/Shuvomoy/NExOS.jl`

Languages

Fluent in
English, Bengali, Hindi, Urdu
Proficient in
`Julia, C, C++, MATLAB, Mathematica`

Other

I enjoy playing cricket, reading novels, cooking, and blogging at `https://shuvomoy.github.io/blogs/`.

Media Coverage

"Risky Giant Steps Can Solve Optimization Problems Faster" August, 2023
by Allison Parshall in *Quanta Magazine*
I was interviewed and quoted in the article along with my paper being cited as the main inspiration for the discovery of long step gradient descent by Ben Grimmer. Also publicized in the *Nautilus Quarterly Magazine* and in the Chinese magazine *Heart of the Machine*.
URL: `https://www.quantamagazine.org/risky-giant-steps-can-solve-optimization-problems-faster-20230811/`